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Article *in* IET Radar, Sonar & Navigation · December 2020

DOI: 10.1049/iet-rsn.2020.0133

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Costas-code-based radar waveform design using adaptive weights with target scattering coefficients and optimal variable time spacing with improved ambiguity function

ISSN 1751-8784
Received on 26th March 2020
Revised 19th August 2020
Accepted on 24th August 2020
E-First on 28th September 2020
doi: 10.1049/iet-rsn.2020.0133
www.ietdl.org

Ramez Eizdashire Ali Deep[1] ✉, *Radwan Kastantin*[1]

[1]*Department of Communication, Higher Institute for Applied Sciences and Technology, Damascus, Syria*

✉ *E-mail: ramez.alideep@hiast.edu.sy*

Abstract: Costas signals are widely used in modern radar systems as frequency coded. The Costas-code-based waveform can achieve a nearly thumbtack shape of the ambiguity function (AF). However, the highest AF's sidelobe levels (SLLs) of Costas waveform equals just 1/***N*** times of its main lobe level height, where ***N*** is the length of the Costas code. In this study, the authors suggest a generalised Costas waveform as a 'burst' of weighted pulses having variable time spacing between them, calling it as weighted variable time spacing Costas (WVTSC). They computed the AF analytical formulas of the designed WVTSC. They developed an adaptation model to adapt the WVTSC to target scattering coefficients, using the interior-point algorithm. The adaptive WVTSC yields much better Doppler cut and improved delay cut in the main lobe area of the AF, in the sense of SLL reduction. Moreover, they constructed an optimisation model resulting in optimal variable time spacing between the sub-pulses. The optimised WVTSC also yields considerable improvement in the AF's recurrent SLLs, using the genetic algorithm. The adapted optimised WVTSC design is proven, through simulation, to have better sidelobe performance than that of the initial waveform, without influencing the original delay-Doppler resolution, and without need to increase ***N***.

1 Introduction

Searching for a practical solution to the waveform synthesis problem is the main goal in radar system design. The waveform controls most system requirements such as the resolution and detection performance. The ambiguity function (AF) concept measures the uncertainty in distinguishing the returning echoes simultaneously. The AF had been referred to as a measure of performance in radar waveform design [1]. Wilcox proposed a solution to the problem by analytically defining the desired ambiguity surface [2]. Wilcox's solution is not practical in any radar application, because in the set of all possible two-dimensional functions AFs are rare. Usually, the practical design of the waveform supposes a desired shape of the ambiguity surface instead of analytical forms, e.g. a thumbtack AF [3–6]. A thumbtack shape had been always considered attractive for numerous radar waveforms. That is because a thumbtack shape provides high delay-Doppler resolution simultaneously. Nevertheless, that shape is useless for waveforms desired to be Doppler tolerant. However, Sussman [7] proposed an approach to the design problem in the sense of least-squared (LS) error to approximate the desired ambiguity surface. Sussman's optimisation procedure runs on the entire delay-Doppler plane. Unfortunately, his optimised waveform cannot be practically suitable for almost every application. Hence, a new extension of Wilcox's method was suggested by the authors in [8, 9]. Their optimisation procedure can be restricted over a limited region in the delay-Doppler plane. Consequently, the designers can focus either on the main lobe or on sidelobe sub-regions, or both, of the AF. We will use this idea in our design.

On the other hand, the designer seeks to have a high-delay-Doppler resolution. Thus, both short pulse time width and small bandwidth (BW) are needed. That is accomplished with the AF having a shape of a thumbtack, where the intervals of both delay and Doppler defining the area of the main lobe are small. However, the need for short time width conflicts with the need of transmitting high energy to get good detection performance. This is because the required energy is achieved by transmitting a long pulse time width instead of using high amplitudes. High amplitudes are constrained by the peak power limitation criteria of the transmitter hardware. To overcome that confliction, the pulse compression technique is a useful solution. This technique allows us to transmit a long pulse time width, and at the same time, the pulse is temporally compressed to achieve high delay resolution.

Pulse compression waveforms can be categorised into two main classes, phase codes and frequency codes [3]. In this study, our waveform optimisation will be based on Costas frequency codes due to the two main reasons. First, Costas codes achieve approximately a near shape of thumbtack, and hence high-delay-Doppler resolution. Second, the Costas sequence of length N yields an AF at zero Doppler cut with sidelobe levels (SLLs) of maximum height $1/N$ times of its main lobe level height. A main drawback of Costas signals is that much better detection performance requires N to be increased [10]. Thus, we want to take the advantage of the high resolution of Costas codes in our design. Then, we aim to decrease the SLLs without the need to increase N.

We present the main idea of our design as follows. Woodward indicated in [1] that the ambiguity quantity in distinguishing multiple targets, for fixed signal-to-noise ratio (SNR), depends on both the probability distribution of the delay shift values τ in the waveform time width T and the probability distribution of the Doppler shift values f_{d} in the waveform bandwidth (BW) B. Precisely, it depends on the joint probability distribution of τ and f_{d}. Moreover, the product TB determines the number of possible resolvable values of τ and f_{d} in the intervals $[-T, +T]$ and $[-B, +B]$, respectively [1]. However, there is a lack of prior knowledge about the possible values of both τ and f_{d}. As a consequence, we assume, by analogy, that there is a map function described as an association of the probability distribution of optimised variable time spacing (VTS) designed between sub-pulses of a Costas sequence to the probability distribution of possible τ values. In addition, we assume that there is another map function described as an association of the probability distribution of possible f_{d} values to the probability distribution of Costas hopping sequence used. It seems to us that this suggestion is reasonable. It would represent a point of view in guessing the target information in terms of modelling optimisation procedures for some parameters.

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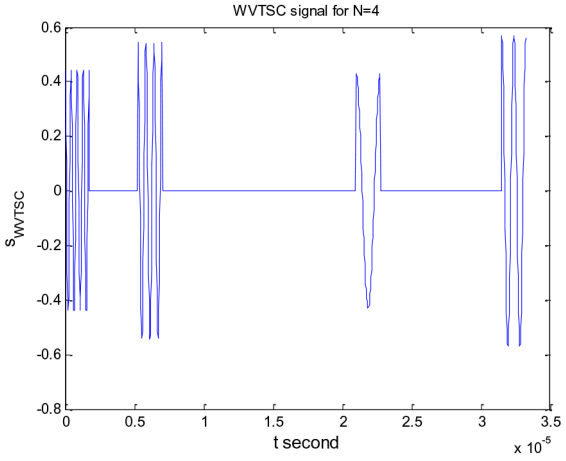


Fig. 1 *Example of the WVTSC burst model*

In this study, we propose an adaptive and optimal waveform design to improve SLL characteristics of the AF at the zero-Doppler cut. The adaptive procedure is based on incorporating scattering coefficients of the target into the AF [11]. We had recently developed an optimisation approach to adaptively design a weighted Costas waveform without applying VTS [12]. We considered that the received signal could depend on the scattering coefficients of the target. We required that the weaker scatterer should be supplied with more energy than the stronger one, without affecting the resolution. We found that the minimisation procedure used could redistribute the signal energy at target scatterers. Our previous work yielded much better auto-correlation function (ACF) in the sense of decreasing SLL, but just in the main lobe area. In this study, we also suggest to incorporate an optimal VTS technique to the original Costas model. We aim to decrease the SLL also in the recurrent sidelobe area of the AF at zero-Doppler cut. By using fixed Golomb ruler (GR) and Costas array as VTS methods, a good reduction of recurrent sidelobe can be achieved [13]. However, no impact of applying those methods was found in the main lobe area. In addition, neither AF mathematical expression of VTS-based Costas was computed nor optimal VTS was considered in [13]. Therefore, we will adjust the models proposed in [12, 13] by incorporating them into a single model. The authors in [14] proposed an optimised waveform for sonar application. Our VTS optimisation approach is quasi similar, but for the radar case. Our main contribution is that we propose a new waveform model aggregating various considerations and assumptions mentioned above. The results obtained will make the proposed model reasonable and technically sound.

The structure of this paper is as follows. In Section 2, we suggest a generalised Costas chirp waveform as a train or a 'burst' of weighted sub-pulses having VTS between them. We consider that each pulse width in the 'burst' equals the original sub-pulse width of Costas. In Section 3, we derive the AF analytical formula of the weighted variable time spacing Costas (WVTSC). The resultant formula allows us to separate it into two parts, the main lobe part and the sidelobe part. In Section 4, we propose a sequential optimisation, first to obtain adaptive weights, then optimal VTS marks between the sub-pulses of the 'burst'. In Section 5, we present the simulation results of the proposed model and discuss them. Finally, in Section 6, we conclude the paper.

2 Waveform model

2.1 Applying VTS to Costas waveform model

The VTS concept between pulses was used in moving target indicator (MTI) radar that is usually called staggered pulse repetition frequency (PRF)-based radar. A Doppler adaptive MTI filter was designed with a variable pulse repetition period, i.e. VTS between the pulses in the train [15]. The patent in [16] presented a method for positioning pulses in time using time hopping codes (THCs) that provided spectral shaping for ultra-wide-band radar. The THC yielded interference rejection and suppression. The THC distribution was also able to improve the spectral properties of the pulse trains. Now, let us consider Costas code whose hopping sequence is $\boldsymbol{c} = [c_0, c_1, \ldots, c_{N-1}]^{\mathrm{T}}$ with N active subcarriers $\{f_n\}_{0 \le n \le N-1}$ each of duration t_b s and equally frequency spaced by δf Hz, where $f_n = c_n/t_b$ Hz and $\delta f \times t_b = 1$. The overall BW of the Costas signal is $B = N\delta f$ Hz, and its pulse duration is $T = Nt_b$ s [10]. Then, the complex envelope of a Costas pulse can be represented as

$$s_{\mathrm{Costas}}(t) = (1/\sqrt{Nt_b}) \sum_{n=0}^{N-1} s_n(t - nt_b), \tag{1}$$

where $s_n(t) = \mathrm{e}^{\mathrm{j}2\pi f_n t}$ for $0 \le t \le t_b$, and zero elsewhere. As the first stage in our waveform design, we apply VTS between sub-pulses of the Costas signal. Then, we can rewrite (1) such that

$$s_{\mathrm{Costas_VTS}}(t) = (1/\sqrt{T_{\max}}) \sum_{n=0}^{N-1} s_n(t - T_n), \tag{2}$$

where each $T_n \in \{T_n\}_{0 \le n \le N-1} = \boldsymbol{T} = [T_0, T_1, \ldots, T_{N-1}]^{\mathrm{T}}$ represents a time mark (a transmission moment) subject to

$$T_0 = 0, \forall i: 2 \le i \le N-1; T_i - T_{i-1} \ge 2t_b, \tag{3}$$

and $T_{\max} = \max(\{T_n\}_{0 \le n \le N-1}) + t_b$ represents the time duration of sending all sub-pulses of $s_{\mathrm{Costas_VTS}}(t)$. Using (3), one can find that $T_{\max} = T_{N-1} + t_b$, because $\{T_n\}_{0 \le n \le N-1}$ is monotonically increasing. For example, we can set up the time marks by choosing a GR as a vector $\boldsymbol{GR} = [0, 1, 4, 6]$ for $N = 4$ [17, 18]. Then, by multiplying $\boldsymbol{GR}$ by $T_r = 3t_b$, we get the VTS marks distribution $\{T_n\}_{0 \le n \le 3} = [0, T_r, 4T_r, 6T_r]$. This setting will be used in selecting the upper bounds of the VTS marks being optimised. In [18], optimal GR up to $N = 27$ marks is illustrated. Each sub-pulse of a VTS Costas signal starts at the moment T_n and ends at $T_n + t_b$, with active subcarrier f_n. Motivationally, the GR-based VTS method was not proven to be an optimum solution to achieve the SLL reduction or not [13]. No one knows the optimal VTS marks. Moreover, there are no analytical rules for now to decide whether the GR-based VTS method is an optimal solution (or the unique one) for the SLL reduction problem, or not.

2.2 Adding weights to VTS waveform model

Employing multiple frequencies makes the target scattering centres differently resonate at each active subcarrier. We aim to demonstrate effects of target impulse response (TIR) on the AF. Let $\boldsymbol{a} = [a_0, a_1, \ldots, a_{N-1}]^{\mathrm{T}}$ be generally a vector of complex weights transmitted over different subcarriers subject to $\sum_{n=0}^{N-1} |a_n|^2 = 1$ to achieve normalised signal energy. We model $\boldsymbol{a}$ in (2) such that

$$s_{\mathrm{WVTSC}}(t) = (1/\sqrt{T_{\max}}) \sum_{n=0}^{N-1} a_n s_n(t - T_n), \tag{4}$$

where $(.)_{\mathrm{WVTSC}}$ refers to the proposed WVTSC model. The quantity $\boldsymbol{a}$ can be adapted with an estimation of TIR. Fig. 1 illustrates an example of the WVTSC burst defined in (4), where we choose $t_b = 1.75\,\mu$s, $T_r = 3t_b$, $\boldsymbol{T} = \{T_n\}_{0 \le n \le 3} = T_r\boldsymbol{GR} = [0, T_r, 4T_r, 6T_r]$, $\boldsymbol{c} = [4, 3, 1, 2]$, $\boldsymbol{GR} = [0, 1, 4, 6]$, and $\boldsymbol{a} = [0.4419,\ 0.5433,\ 0.4299,\ 0.5699]$.

The proposed WVTSC is no longer called a Costas waveform or a train of original Costas pulses. It represents instead a burst-like waveform. Its subcarriers are designed based on the Costas code. Its pulses are time spaced according to specific moments during transmission. The WVTSC can also be thought of as a train of pulses. Its amplitudes have different levels and can be adapted with target scattering coefficients. The original Costas waveform includes neither time spaces between its sub-pulses nor weighting. In this paper, we aim to get a WVTSC burst such that it consists of

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a bunch of adaptively-weighted optimally-time-spaced pulses. However, one can think of other methods to distribute $\{f_n\}$ and $\{T_n\}$, e.g. GR and THC, respectively. In general, the distributions of $\{f_n\}$, $\{T_n\}$, and $\{a_n\}$ altogether represent the main idea in our design. The WVTSC can be extended with other distributions.

2.3 Received WVTSC model

In this subsection, we assume that the transmitted radar signal is narrowband. We ignore noise in the radar environment. We consider a far-field point target moving at radial velocity v (m/s) with respect to the radar location. The target distance is R metres far away from the radar. We also suppose that the velocity v is constant during $T_{\max}$. Hence, we get $\tau = 2R/v_c$ and $f_d = 2v/\lambda$, respectively, where $\lambda = v_c/f_c$ is the wavelength, v_c is the speed of light, and f_c is the carrier frequency. Then, by following the approximation in [2], the received WVTSC can be represented as a shifted time-frequency replica of the transmitted one such that

$$y^{(1)}_{\text{received}}(t) = s_{\text{WVTSC}}(t-\tau)\mathrm{e}^{\mathrm{j}2\pi f_d t}. \tag{5}$$

Substituting (4) into (5), we get

$$y^{(2)}_{\text{received}}(t) = (1/\sqrt{T_{\max}})\sum_{n=0}^{N-1} a_n s_n(t-\tau-T_n)\mathrm{e}^{\mathrm{j}2\pi f_d t}. \tag{6}$$

We want to incorporate the target scattering coefficients in our model. Let $\boldsymbol{x} = [x_0, x_1, \ldots, x_{N-1}]^{\mathrm{T}}$ be generally a vector of complex scattering coefficients of the target at different subcarriers. We assume that $\boldsymbol{x}$ represents a priori knowledge from TIR estimation. The received WVTSC can be written as

$$y_{\text{received}}(t) = \frac{1}{\sqrt{T_{\max}}}\sum_{n=0}^{N-1} x_n a_n s_n(t-\tau-T_n)\mathrm{e}^{\mathrm{j}2\pi f_d t}. \tag{7}$$

As a numerical example for simulation and to simplifying the signal processing of our model, we will use Rayleigh distribution as an estimation of $\boldsymbol{x}$. In general, $\boldsymbol{x}$ can be drawn from any other suitable distribution without restriction. The frequency is not constant for the pulses in the WVTSC burst, so it can be assumed that modelling using $\boldsymbol{x}$ is reasonable. A priori knowledge about $\boldsymbol{x}$ is one of the basic assumptions in the WVTSC model. The proposed WVTSC optimisation model is based on the pre-estimation of the TIR.

The backscatter of the target (i.e. TIR) is considered as a recorded complex quantity in the frequency domain. The inverse Fourier transform of the TIR shows peaks (or doublets) at supposed scattering centres. TIR estimation demands exact knowledge of both the magnitude and phase of the echo return, which is not the case in some practical implementations. There is a lack of knowledge about the absolute phase of the target echo. Another lack of knowledge about echo's exact zero-time reference is also faced here. Moreover, phase information total loss of the target's echo exhibits ambiguity in the sense of TIR estimation.

A classical work [19] proposed an approach to scattering centre extraction. This approach was developed based on a model derived from the geometric theory of diffraction. An estimation algorithm for that model was developed and shown to have excellent performance in the Prony algorithm. In [20], a phase retrieval algorithm of sparse signals was developed to estimate the TIR. The authors considered the case in which the echo signal was known to be sparse. The proposed algorithm did not require matrix lifting. Hence, it is potentially appropriate for large scale problems of echo's phase retrieval. The author in [21] is concerned with the total loss of phase information ambiguity, where only the magnitude of the radar return is available. TIR was estimated using only the magnitude of the backscatter. His algorithm was verified for synthetic targets. It was also proven to be applied to real targets using real returns data of commercial target models recorded in a compact range. In our study, we focus on a presumed distribution of target echo magnitudes. When the transmitted weight of each pulse in the WVTSC burst is a_n, the scattering phenomena will multiply a_n by a coefficient x_n such that the magnitude of the returning echo will be $a_n x_n$ as mentioned in (7).

3 Deriving analytical formula of the AF for a single WVTSC waveform model

3.1 AF model of a single WVTSC burst

In this work, we interpret the AF as a matched filter (MF) response including TIR estimation. We define the narrowband AF (NAF) of a single WVTSC waveform as following:

$$\chi_{\text{MF}}(\tau, f_d) \triangleq \int_{-\infty}^{+\infty} s_{\text{WVTSC}}(t) y^*_{\text{reveived}}(t)\mathrm{d}t, \tag{8}$$

where $(.)^*$ denotes the complex conjugate operator. Substituting both (4) and (7) into (8), we obtain

$$\chi_{\text{MF}}(\tau, f_d) = \frac{1}{T_{\max}}\sum_{n_1=0}^{N-1}\sum_{n_2=0}^{N-1} x^*_{n_2} a^*_{n_2} a_{n_1}\int_{-\infty}^{+\infty} s_{n_1}(t-T_{n_1}) s^*_{n_2} \left(t-\tau-T_{n_2}\right)\mathrm{e}^{-\mathrm{j}2\pi f_d t}\mathrm{d}t. \tag{9}$$

Here, we define the cross-AF (CAF) between $s_{n_1}(t-T_{n_1})$ and $s_{n_2}(t-T_{n_2})$ as following:

$$\chi_{s_{n_1}s_{n_2}}(\tau, f_d) = \frac{1}{t_b}\int_{-\infty}^{+\infty} s_{n_1}(t-T_{n_1}) s^*_{n_2}(t-\tau-T_{n_2})\mathrm{e}^{-\mathrm{j}2\pi f_d t}\mathrm{d}t. \tag{10}$$

When $n_1 = n_2 = n$, we define $\chi_{s_n s_n}(\tau, f_d)$ as the auto-AF (AAF) of $s_n(t-T_n)$. Hence, the general formula of (9) can be rewritten as

$$\chi_{\text{MF}}(\tau, f_d) = \frac{t_b}{T_{\max}}\sum_{n_1=0}^{N-1}\sum_{n_2=0}^{N-1} x^*_{n_2} a^*_{n_2} a_{n_1}\chi_{s_{n_1}s_{n_2}}(\tau, f_d). \tag{11}$$

Equation (11) suggests a modified Woodward's definition of the AF. The resulted MF response includes the scattering coefficients of the target $\boldsymbol{x}$ and the weighting coefficients $\boldsymbol{a}$. This representation is more suitable when interpreting $\boldsymbol{x}$ as a target 'fingerprint'. Equation (11) can be considered as an increment improvement over representing the target echo. It allows us to perform an adaptive WVTSC design such that $\boldsymbol{a}$ can be adaptively selected according to $\boldsymbol{x}$. Moreover, (11) can be simplified such that we can separate the total ambiguity into two ambiguous quantities. The first is the main lobe part of the NAF. The second represents the sidelobe part. Hence, we can separately control the SLL reduction in both the main lobe and the recurrent sidelobe areas. This is done by searching the corresponded adaptive weights and optimal VTS. We simplify (11) (see the proof in the Appendix) such that

$$\chi_{\text{MF}}(\tau, f_d) = \chi^{(\text{ml})}_{\text{MF}}(\tau, f_d) + \chi^{(\text{sl})}_{\text{MF}}(\tau, f_d), \tag{12}$$

where $\chi^{(\text{ml})}_{\text{MF}}(\tau, f_d)$ is the main-lobe part and $\chi^{(\text{sl})}_{\text{MF}}(\tau, f_d)$ is the recurrent sidelobe part of the NAF. When $|\tau| \le t_b$, we get

$$\chi^{(\text{ml})}_{\text{MF}}(\tau, f_d) = (1/T_{\max})\sum_{n=0}^{N-1} x^*_n |a_n|^2 (t_b - |\tau|)\text{sinc}(\pi f_d(t_b-|\tau|))\mathrm{e}^{\mathrm{j}\pi\kappa_n(\tau, f_d)}, \tag{13}$$

and $\chi^{(\text{ml})}_{\text{MF}}(\tau, f_d) = 0$ elsewhere, where we put $\kappa_n(\tau, f_d) = -f_d(2T_n + t_b + \tau) + 2f_n\tau$. When $\left|\tau - (T_{n_1} - T_{n_2})\right| \le t_b$, we get

(see (14))

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$$\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, f_{\mathrm{d}}) = (1/T_{\max}) \sum_{n_1=0}^{N-1} \sum_{\substack{n_2=0 \\ n_2 \neq n_1}}^{N-1} x_{n_2}^{*} a_{n_2}^{*} a_{n_1}\left(t_b - \left|\eta_{n_1 n_2}(\tau)\right|\right) \mathrm{sinc}\left(\pi\alpha\left(t_b - \left|\eta_{n_1 n_2}(\tau)\right|\right)\right) \mathrm{e}^{\mathrm{j}\pi\zeta_{n_1 n_2}(\tau, f_{\mathrm{d}})}, \tag{14}$$

and $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, f_{\mathrm{d}}) = 0$ elsewhere, where we put $\eta_{n_1 n_2}(\tau) = \tau - \left(T_{n_1} - T_{n_2}\right)$, $\zeta_{n_1 n_2}(\tau, f_{\mathrm{d}}) = \alpha\left(t_b + \eta_{n_1 n_2}(\tau)\right) + 2 f_{n_2} \eta_{n_1 n_2}(\tau) - 2 f_{\mathrm{d}} T_{n_1}$, and $\alpha = f_{n_1} - f_{n_2} - f_{\mathrm{d}}$.

The proposed NAF is a function not only of τ and f_{d} but also of both $\boldsymbol{a}$ and $\boldsymbol{x}$. Thus, we normalise it by computing the ratio $\left|\chi_{\mathrm{MF}}(\tau, f_{\mathrm{d}}, \boldsymbol{a}, \boldsymbol{x}) / \chi_{\mathrm{MF}}(0, 0, \boldsymbol{a}, \boldsymbol{x})\right|$ in the simulation. The purpose here is to study the volume of the NAF surface as a unitary quantity. On the other hand, we will call the magnitude $\left|\chi_{\mathrm{MF}}(\,.\,)\right|$ as an ambiguity surface, and the integral $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \left|\chi_{\mathrm{MF}}(\tau, f_{\mathrm{d}})\right|^2 \mathrm{d}\tau \mathrm{d} f_{\mathrm{d}}$ as volume under the surface.

Currently, we are working on another topic to study the issue of sending multi-PRF WVTSC waveforms with a medium PRF. A medium PRF is necessitating the resolution of range and Doppler ambiguities. Hence, for any joint optimisation of several WVTSC waveforms, constraints on $T_{\max}$ should be added to the optimisation. That is because $T_{\max}^{(i)}$ of each ith waveform defines the blind range (among other things). Moreover, $T_{\max}$ would fluctuate unpredictably. To deal with the optimisation of multi-medium-PRF WVTSC waveforms, M of N schedule would be a possible approach following [22]. Unambiguous range and velocity may be decoded through a comparison of the ambiguous target data received in a minimum number, M, PRFs [22]. The blindness regions require that a radar must alter its operation over several N coherent bursts of PRFs. This mode is to recover sufficient data in the requisite M multi-PRF waveforms to resolve ambiguity. To make our scheme compatible with the optimisation procedure of [22], the total time required for transmission of the waveforms must be less than the dwell time used, i.e. ($\sum_i T_{\max}^{(i)} <$ dwell time) as an added constraint.

3.2 WVTSC correlation property and the corresponded Doppler intolerance drawback

In the absence of Doppler shift ($f_{\mathrm{d}} = 0$), the AAF of (10) represents the ACF of $s_n(t - T_n)$, i.e. $\chi_{s_n s_n}(\tau, 0)$, and the CAF of (10) represents the cross-correlation function (CCF) between $s_{n_1}\left(t - T_{n_1}\right)$ and $s_{n_2}\left(t - T_{n_2}\right)$, i.e. $\chi_{s_{n_1} s_{n_2}}(\tau, 0)$. As a consequence, $\chi_{\mathrm{MF}}(\tau, 0)$ derived in (12) will embody the aggregate correlation properties, i.e. both the auto-correlation $\chi_{\mathrm{MF}}^{(\mathrm{ml})}(\tau, 0)$ and the cross-correlation $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, 0)$ properties. The main lobe part $\chi_{\mathrm{MF}}^{(\mathrm{ml})}(\tau, 0)$ is desired to be a Dirac delta function by performing an adaptation process. The recurrent sidelobe part $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, 0)$ can be viewed as the interference in $\chi_{\mathrm{MF}}(\tau, 0)$, and hence it is desired to be zero. The main goal of both the adaptation and optimisation processes is to achieve the ideal $\chi_{\mathrm{MF}}(\tau, 0)$. This firstly means that the SLL of $\chi_{\mathrm{MF}}^{(\mathrm{ml})}(\tau, 0)$ in the main lobe area should be suppressed by searching the adaptive weights. It also means that the SLL of $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, 0)$ in the recurrent sidelobe area should be reduced by searching the optimal time marks. These operations are supposed to be performed at the transmitter side. The performance of the optimised WVTSC is checked by those SLL. Those levels should be as low as possible to minimise the false alarm probability.

The peak SLL (PSL) is usually used as metrication of the SLL. For instance, the PSL is affected in the presence of Doppler shift ($f_{\mathrm{d}} \neq 0$). Hence, if a good detection ability of both stationary and moving targets is desired to be achieved, the PSL should be low for all f_{d} values. A moving target's peak power is affected if $f_{\mathrm{d}} \neq 0$. The degradation in the target's peak power for increasing relative velocities, especially for very fast-moving targets, implies SNR loss resulting from an uncompensated Doppler shift at the MF output. A waveform exhibiting a very narrow Doppler peak would result in strongly degraded detection performance for a very fast moving target. Hence, if $f_{\mathrm{d}} \neq 0$, the good correlation properties of the waveform can be broken regarding the SNR loss. This effect is called the Doppler intolerance of radar waveforms. It is a serious problem for most pulse-Doppler radar systems that perform first pulse compression and second fast Fourier transform-based Doppler estimation. This is true of the presented optimised WVTSC. That is because the resulted NAF is here desired to have a nearly thumbtack shape in the entire delay-Doppler plane.

Consequently, Doppler intolerance drawback implies imperfections in the correlation properties of $\chi_{\mathrm{MF}}(\tau, 0)$, i.e. the ideal properties of both the ACF and the CCF of the optimised WVTSC are lost. Returning to the two functions $\kappa_n(\tau, f_{\mathrm{d}})$ and $\zeta_{n_1 n_2}(\tau, f_{\mathrm{d}})$ formulas defined in (13) and (14), respectively, one can see that phase shifts are produced when $f_{\mathrm{d}} \neq 0$ (with $\tau \neq 0$) within each sub-pulse of the WVTSC burst. Those intra-pulse phase shifts are what will break both $\chi_{\mathrm{MF}}^{(\mathrm{ml})}(\tau, 0)$ and $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, 0)$ desired properties of the pulse sequences in a single WVTSC burst. Furthermore, in the situation of transmitting consecutive WVTSC bursts, inter-pulse phase shifts can also break those correlation properties. Thus, Doppler phase shifts make the SLL of the $\chi_{\mathrm{MF}}(\tau, f_{\mathrm{d}})$ increase. Then, these phase shifts impair the detection ability for moving targets. However, the PSL being as function of the Doppler shift can be used as a measurable indicator for the Doppler intolerance drawback. This means that if $\chi_{\mathrm{MF}}(\tau, 0)$ is desired to be a Dirac delta function with zero SLL, the PSL at $f_{\mathrm{d}} = 0$ should approximately equal to zero. On the other hand, when the absolute value of f_{d} increased, the corresponding PSL results should be much higher than that at $f_{\mathrm{d}} = 0$. Thus, the good Doppler tolerance of the proposed WVTSC can be indicated if the PSLs computed at each non-zero f_{d} were equal to the PSL computed at $f_{\mathrm{d}} = 0$.

There are three possible solutions to mitigate the Doppler intolerance effect. The first is to consider ideal correlation properties for given Doppler frequencies ($\left[f_{\mathrm{d}}^{\min}, f_{\mathrm{d}}^{\max}\right]$). This could be done by adding two constraints to the proposed WVTSC optimisation. Precisely, it is to force both the AAF and the CAF of the WVTSC to be Dirac delta function and zero-function, respectively. This solution requires that the desired NAF shape should be a constant amplitude ridge (e.g. kept above 0.5) along with the zero range in the given Doppler BW, instead of using a thumbtack shape. This solution should be implemented at the transmitter side. Hence, more computation power is required at the transmitter since both weights and time marks of the WVTSC are optimised. At first glance, this solution does not seem very good since the resulted optimised WVTSC will be only tolerant at limited Doppler shifts. Moreover, it is computationally complex since it is implemented at the transmitter side along with searching both adaptive weights and optimal VTS for the WVTSC. The second solution is a multi-hypothesis Doppler shift matched filtering performed at the receiver side [23], but it is generally known that most of deployed radar systems only assert that targets have a null Doppler shift. The third solution is to compensate for the Doppler shifts by utilising signal processing techniques also at the receiver side. The common approach here is to use a bank of Doppler-matched filters. A more developed approach based on that bank of such filters was recently proposed in [24]. It applies the Doppler compensation technique based on the weighting method to compensate for the intra-pulse Doppler phase shifts. It seems to us that the third solution is more attractive to obtain the Doppler resilience of the proposed WVTSC. That is because phase shifts are also introduced in the WVTSC model. In addition, it could be

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implemented at the receiver side, so we can distribute the signal processing operations into the two sides of the radar. The challenge with this compensation method is that the Doppler phase shifts over several pulses of the WVTSC burst instead of a single pulse to be compensated. We can exactly follow the model proposed in [24] to overcome that, since our model has a quasi-similar situation. This is left for a future work. However, we will give an estimate of the degree of the Doppler intolerance of the optimised WVTSC in Section 5. This would give readers an idea about the Doppler intolerance effect of the WVTSC.

4 Adaptation and optimisation approach for the waveform model

4.1 Minimisation problems formulation

We introduce an adaptation procedure to control the SLL in the NAF main lobe area. An optimisation procedure to control the SLL in the recurrent sidelobe areas is also proposed. We develop two types of minimisation procedures. The first is to get adaptive $\boldsymbol{a}$, and the second is to get optimal $\boldsymbol{T}$. The resultant adaptive $\boldsymbol{a}$ will redistribute the signal energy such that the weakest target scatterer $[\min(\boldsymbol{x})]$, e.g. x_2, requires more energy to be supplied from the corresponded a_2. This means that the magnitude of the adaptive weight a_2 should result being the largest, and vice versa. The resultant optimal $\boldsymbol{T}$ will rearrange WVTSC pulses in $[0, T_{\max}]$. We measure the ability of the adaption in redistributing energy by a ratio $RQ = \mathrm{var}(|\langle \boldsymbol{a}_{\text{adaptive}}, \boldsymbol{x}\rangle|)/\mathrm{var}(|\langle \boldsymbol{a}_{\text{initial}}, \boldsymbol{x}\rangle|)$. The adaption has a strong ability to redistribute the energy for $RQ \ll 1$, but the weak ability for $RQ \simeq 1$. $\langle \boldsymbol{a}, \boldsymbol{x}\rangle$ is the common scalar product of the two vectors $\boldsymbol{a}, \boldsymbol{x}$, and $\mathrm{var}(.)$ is the variance operator. Being $RQ \ll 1$ is equivalent to getting $\mathrm{var}(|\langle \boldsymbol{a}_{\text{adaptive}}, \boldsymbol{x}\rangle|)$ very small than $\mathrm{var}(|\langle \boldsymbol{a}_{\text{initial}}, \boldsymbol{x}\rangle|)$. This means that RQ is closer to zero, and the amplitude of the received echo is much more constant over the reception time. $\boldsymbol{a}_{\text{initial}}$ represents the initial weights, and we will consider that $\boldsymbol{a}_{\text{initial}}(n) = 1/\sqrt{N},\ 0 \le n \le N-1$.

For the adaptation procedure, we assume a desired ambiguity surface defined over a desired limited sub-region of (τ, f_{d})-plane. This assumption is totally based on the works [8, 9]. In our work, we focus on the thumbtack shape of the AF. To simplify the signal processing, we suppose the desired ambiguity surface such that $|\chi_{\text{Opt}}(\tau, f_{\mathrm{d}})| = 1$, for $(\tau, f_{\mathrm{d}}) = (0, 0)$, and zero elsewhere. This profile has zero-main lobe width. However, another thumbtack profile can also be used according to the designer, e.g. having a finite main lobe width. To achieve adaptive $\boldsymbol{a}$, we minimise the LS error between the volume of $|\chi_{\mathrm{MF}}(.)|$ and that of $|\chi_{\text{Opt}}(.)|$. For both the adaptation and optimisation procedures, we assume limited sub-regions Ω in the (τ, f_{d})-plane. We will use the sub-regions of rectangular shape. Other geometries can be used for Ω, e.g. circular. The Ω selection is up to the designer.

4.2 Adaptation procedure for the WVTSC

The first requirement of WVTSC design is to get considerable SLL reduction in the main lobe area ($|\tau| \le t_b$). Therefore, we suggest a minimisation problem corresponding to find the adaptive weights $\boldsymbol{a}_{\text{adaptive}}$ as follows:

$$\left[\boldsymbol{a}_{\text{adaptive}}(\boldsymbol{x})\right] = \arg\min_{\boldsymbol{a}}\left\{\int_{-t_b}^{+t_b}\int_{-1/t_b}^{1/t_b}\left||\chi_{\mathrm{MF}}| - |\chi_{\text{Opt}}|\right|^2 \mathrm{d}\tau\mathrm{d}f_{\mathrm{d}}\right\}, \quad (15)$$

subject to $\sum_{n=0}^{N-1}|a_n|^2 = 1$ with $\forall n; \exists \varepsilon_1 > 0: |a_n|^2 \ge \varepsilon_1$, and $\exists \varepsilon_2 > 0: \mathrm{var}(\boldsymbol{a}) \le \varepsilon_2$, where ε_1 and ε_2 are small positive quantities relatively close to zero. $\sum_{n=0}^{N-1}|a_n|^2 = 1$, $\mathrm{var}(\boldsymbol{a}) \le \varepsilon_2$, and $|a_n|^2 \ge \varepsilon_1$ represent non-linear equality and inequality constraints. The inequality $|a_n|^2 \ge \varepsilon_1$ is to ensure transmitting all of the burst pulses. The $\mathrm{var}(\boldsymbol{a}) \le \varepsilon_2$ is to take into consideration the transmitter power limits [5], such that we could accept the relatively small variations in the WVTSC amplitude. $\boldsymbol{a}_{\text{adaptive}}$ clearly adapts to the target scattering coefficients $\boldsymbol{x}$. We choose $[-t_b, +t_b]$ as the integration domain for τ variations. This domain defines the main lobe area. It is very important to state here that, during the adaptation process, not only keeping the overall energy of the WVTSC equal to unity, but also the amplitude of all its pulses must be kept constant. This is so required to guarantee that all pulses have energy radiated equal to unity (or kept constant) when the optimisation is performed. It also means that there is no pulse variation in amplitude as the optimisation progresses.

At the beginning of the optimisation process, if the volume under $|\chi_{\text{Opt}}|$ is much smaller than that under $|\chi_{\mathrm{MF}}|$, the next problem simplifies (15) (with the same constraints)

$$\left[\boldsymbol{a}_{\text{adaptive}}(\boldsymbol{x})\right] = \arg\min_{\boldsymbol{a}}\left\{\int_{-t_b}^{+t_b}\int_{-1/t_b}^{1/t_b}|\chi_{\mathrm{MF}}|^2 \mathrm{d}\tau\mathrm{d}f_{\mathrm{d}}\right\}, \quad (16)$$

or

$$\left[\boldsymbol{a}_{\text{adaptive}}(\boldsymbol{x})\right] = \arg\min_{\boldsymbol{a}}\left[\sum_{n_1=0}^{N-1}\sum_{n_2=0}^{N-1}\sum_{n_3=0}^{N-1}\sum_{n_4=0}^{N-1}\left\{a_{n_1}a_{n_2}^{*}a_{n_3}^{*}a_{n_4}x_{n_3}^{*}x_{n_4}\right.\right.$$
$$\left.\left.\int_{-t_b}^{+t_b}\int_{-1/t_b}^{1/t_b}\chi_{s_{n_1}s_{n_2}}(\tau, f_d)\chi_{s_{n_3}s_{n_4}}^{*}(\tau, f_{\mathrm{d}})\mathrm{d}\tau\mathrm{d}f_{\mathrm{d}}\right\}\right],$$

where $\chi_{s_{n_1}s_{n_2}}(\tau, f_{\mathrm{d}})$ satisfies (10). In the Appendix, we show that the general formula of (10) satisfies

- Case 1: when $n_1 = n_2 = n$, $|\tau| \le t_b$, we get

$$\chi_{s_n s_n}(\tau, f_{\mathrm{d}}) = \frac{t_b - |\tau|}{t_b}\mathrm{sinc}(\pi f_{\mathrm{d}}(t_b - |\tau|))\mathrm{e}^{\mathrm{j}\pi\kappa_n(\tau, f_{\mathrm{d}})}, \quad (17)$$

  and zero elsewhere.
- Case 2: when $n_1 \ne n_2$, $|\tau - (T_{n_1} - T_{n_2})| \le t_b$, we get
  (see (18))
  and zero elsewhere.

The sum term in (16) leads to a minimisation problem having fourth order, i.e. if $n_i = n, i = 1, 2, 3, 4$, we have the term $|a_n|^4$. This implies the fourth-order polynomial objective function. The convergence time of (15) and (16) strongly depends on both the area of Ω and the $|\chi_{\text{Opt}}|$ shape. However, assuming that a priori knowledge of $\boldsymbol{x}$ can be guaranteed for the next transmitted WVTSC (estimating $\boldsymbol{x}$ from the previous radar dwell), it would be practical to restrict Ω to be smaller. This yields no necessity to consider $|\chi_{\text{Opt}}|$ shape. Hence, (16) can be used instead of (15) to speed up signal processing. To further speed up the minimisation process, one can use just the main lobe part of the NAF and replace $|\chi_{\mathrm{MF}}(.)|$ with $|\chi_{\mathrm{MF}}^{(\mathrm{ml})}(.)|$ in (16). Thus, the problem (16) (with the same constraints) can be simplified as follows:
(see (19))

4.3 Optimisation procedure for the WVTSC

The second requirement of WVTSC design is to get considerable SLL reduction in the recurrent sidelobe areas ($|\tau| > t_b$). Therefore, we suggest a minimisation problem corresponding to finding the optimal positive real-time marks $\boldsymbol{T}_{\text{Optimal}} = \left\{T_n^{\text{optimal}}\right\}_{0 \le n \le N-1}$. This problem corresponds to maximise SLL reduction of each recurrent

$$\chi_{s_{n_1}s_{n_2}}(\tau, f_{\mathrm{d}}) = \frac{t_b - |\eta_{n_1n_2}(\tau)|}{t_b}\mathrm{sinc}\left(\pi\alpha\left(t_b - |\eta_{n_1n_2}(\tau)|\right)\right)\mathrm{e}^{\mathrm{j}\pi\tilde{\zeta}_{n_1n_2}(\tau, f_{\mathrm{d}})}, \quad (18)$$

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sidelobe of the NAF. In other words, the volume under the ambiguity surface corresponding to the recurrent sidelobe part $\left|\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, f_{\mathrm{d}})\right|$ is minimised. The suggested problem can be categorised as multi-objects optimisation with linear constraints such that

$$\left[\boldsymbol{T}_{\mathrm{Optimal}}\right] = \underset{\{T_n\}_{0 \le n \le N-1}}{\operatorname{argmin}} \left\{ \sum_{s=0}^{N-1} \omega_s F_s \right\}, \tag{20}$$

subject to $(T_0 = 0)$, and $\forall s = 2, \ldots, N-1; T_s - T_{s-1} \ge 2t_b$. One can also add constraints on T_{N-1}. The function $F_s = \int_{T_s - t_b}^{T_s + t_b} \int_{-1/t_b}^{+1/t_b} \left|\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}})\right|^2 \mathrm{d}\tau \mathrm{d}f_{\mathrm{d}}$ is the objective of the minimisation procedure (20). Each F_s corresponds to the sth recurrent sidelobe function $\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}})$. The $\{\omega_s\}_s$ are weighting coefficients described in the next subsection. F_s represents volume quantity. One can notice here that $\chi_{\mathrm{MF}}^{(\mathrm{sl})}(\tau, f_{\mathrm{d}}) = \sum_{s=0}^{N-1} \chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}})$. According to (14), we can write, for $T_s - t_b \le \tau \le T_s + t_b$

$$\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}}) = (a_s / T_{\max}) \sum_{\substack{n=0 \\ n \ne s}}^{N-1} x_n^* a_n^* (t_b - |\eta_{sn}(\tau)|) \mathrm{sinc}(\pi\alpha(t_b - |\eta_{sn}(\tau)|)) \mathrm{e}^{\mathrm{j}\pi\zeta_{sn}(\tau, f_{\mathrm{d}})} \tag{21}$$

where $\{a_s\}_{0 \le s \le N-1}$ is the solution of the adaptation problem. The first recurrent sidelobe $(s = 1)$ will be $\chi_{\mathrm{MF}}^{(\mathrm{sl})_1}(\tau, f_{\mathrm{d}})$ for $T_1 - t_b \le \tau \le T_1 + t_b$. The second recurrent sidelobe $(s = 2)$ will be $\chi_{\mathrm{MF}}^{(\mathrm{sl})_2}(\tau, f_{\mathrm{d}})$ for $T_2 - t_b \le \tau \le T_2 + t_b$, and so on. For instance, in the constant time spacing situation, zeros occur between recurrent side lobes. This is due to the constraint proposed in (3). In fact, each $\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}})$ represents a CCF between one pulse of transmitted WVTSC and one pulse of its received replica. In addition, the shape of each $\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(\tau, f_{\mathrm{d}})$ depends on both the initial VTS marks and the original Costas waveform parameters. The initial VTS marks can be computed either by directly using a GR or by using constant time spacing, e.g. $T_n^{\mathrm{initial}} = nT_r$, where $T_r \ge 2t_b$. T_r represents that possible constant time spacing. The main reason behind considering just $\chi_{\mathrm{MF}}^{(\mathrm{sl})_s}(.)$ in formulating (20) is that $\chi_{\mathrm{MF}}^{(\mathrm{ml})}(.)$ does not really play any role in the recurrent SLL areas. However, it should be noted that $\chi_{\mathrm{MF}}^{(\mathrm{sl})_0}(.)$ is defined for $|\tau| \le t_b$ because T_0 is always zero according to constraint (3). $\chi_{\mathrm{MF}}^{(\mathrm{sl})_0}(.)$ plays a basic role just in the main lobe area. Thus, there is no necessity to mention it as a recurrent sidelobe. Nevertheless, $\chi_{\mathrm{MF}}^{(\mathrm{sl})_0}(.)$ is included in (20) because it represents a sum term in (14). In the rest of this article, we consider that there are just $N-1$ recurrent side lobes included in (20) and defined for $t_b \le \tau \le T_{\max} = T_{N-1} + t_b$ (positive τ). For the sake of reducing the computation complexity, it is possible to restrict the number of F_s in (20) to a smaller number, e.g. $0 \le s \le 3$ (three F_s). This might be done since the SLLs of the remained objective functions are relatively low originally and located far away from the main lobe. That would increase the chance to approach an optimal solution much quicker. In Section 5, we will show that $\left[\boldsymbol{T}_{\mathrm{Optimal}}\right]$ marks yield the following two expected main results: (i) recurrent SLL at zero Doppler cut, for $T_s^{\mathrm{optimal}} - t_b \le \tau \le T_s^{\mathrm{optimal}} + t_b$, will be too lower than those for $sT_r - t_b \le \tau \le sT_r + t_b$, (ii) NAF zeros occurring between recurrent sidelobe functions at zero Doppler cut, for optimal time marks, will be reduced or removed.

4.4 Clarifying the algorithms used to solve the minimisation problems

The proposed adaptation problems correspond to find minima of a constrained non-linear multivariable function $f(\boldsymbol{y})$. Each one of them can be considered as a special case of a general minimisation problem specified by

$$\min_{\boldsymbol{y}} f(\boldsymbol{y}) \text{ subject to } \begin{cases} c(\boldsymbol{y}) \le 0, \ \mathrm{ceq}(\boldsymbol{y}) = 0 \\ \boldsymbol{A} \times \boldsymbol{y} \le \boldsymbol{b}, \boldsymbol{Aeq} \times \boldsymbol{y} = \boldsymbol{beq}, \\ \boldsymbol{lb} \le \boldsymbol{y} \le \boldsymbol{ub} \end{cases} \tag{22}$$

where $\boldsymbol{y} \equiv \boldsymbol{a} \equiv \{a_n\}_{0 \le n \le N-1}$ represents the several variables (i.e. a vector) starting at an initial estimate $\boldsymbol{a}_{\mathrm{initial}}$. $f(\boldsymbol{a}) = \int_{-t_b}^{+t_b} \int_{-1/t_b}^{1/t_b} \left\| \chi_{\mathrm{MF}} | - \left| \chi_{\mathrm{Opt}} \right| \right\|^2 \mathrm{d}\tau \mathrm{d}f_{\mathrm{d}}$ is the fitness (objective) function that returns in a scalar. $c(\boldsymbol{y})$ represents the set of non-linear inequality constraints. $\mathrm{ceq}(\boldsymbol{y})$ is the set of non-linear equality constraints. $\boldsymbol{A}, \boldsymbol{b}$ represent the linear inequality constraints ($\boldsymbol{A}$ is a matrix, $\boldsymbol{b}$ is a vector). $\boldsymbol{Aeq}$ and $\boldsymbol{beq}$ represent the linear equality constraints ($\boldsymbol{Aeq}$ is a matrix, $\boldsymbol{beq}$ is a vector). $\boldsymbol{lb}$ and $\boldsymbol{ub}$ are the lower bounds and upper bounds of the several variables of function f ($\boldsymbol{lb}$ and $\boldsymbol{ub}$ are vectors). Equation (22) is generally referred to as constrained non-linear optimisation problem (i.e. non-linear programming) [25, 26]. For our adaptation problems, there are no linear inequality or linear equality constraints. Instead, there are $N+1$ non-linear inequality constraints represented by both $|a_n|^2 \ge \varepsilon_1$ and $\mathrm{var}(\boldsymbol{a}) \le \varepsilon_2$. A non-linear equality constraint is also represented by $\sum_{n=0}^{N-1} |a_n|^2 = 1$. We consider the lower bounds ($\boldsymbol{lb}$ elements) of (15) are zeros, whereas the upper bounds ($\boldsymbol{ub}$ elements) are ones. Numerous algorithms can be used to solve (22), e.g. the interior-point and trust-region-reflective. These algorithms allow to supply a *Hessian multiply* function, which gives the result of a Hessian-times-vector product, without computing the Hessian directly, and hence saving memory. For more extensive description about these two algorithms see [25, 26].

The optimisation problem (20) corresponds to find the minima of multiple functions using the genetic algorithm (GA). The multiple functions here are $\{F_s\}_{0 \le s \le N-1}$. According to Deb [27], multi-objective optimisation (MOO) using evolutionary algorithms implement the GA to minimise a multicomponent objective function represented by $F(\boldsymbol{T}) = \sum_{s=0}^{N-1} \omega_s F_s$. A controlled elitist GA (i.e. a variant of non-dominated sorting GA II) is used to favour individuals with a better fitness value (rank). Hence, these individuals yield to increase the diversity of the population even if individuals are getting a lower fitness value. Therefore, a controlled elitist GA implies maintaining the diversity of the population for convergence to an optimal Pareto front by controlling the elite members of the population as the algorithm progresses [27]. Although (20) has no non-linear constraints, it includes linear inequality and equality constraints represented by $(\forall s = 2, \ldots, N-1; T_s - T_{s-1} \ge 2t_b)$ and $(T_0 = 0)$, respectively. At first glance, one might see that there is no control upon T_{N-1}, the last-time mark that defines $T_{\max}$ proposed in (2). However, a constraint on the first-time mark T_0 is just added. Nevertheless, there are upper bounds $\boldsymbol{ub}$ of (20) for the wanted $\{T_n\}_{0 \le n \le N-1}$, and hence for T_{N-1}, so for $T_{\max}$. These upper bounds are required as an input parameter for running (20). Thus, it can be seen that a constraint is added indirectly to (20) so that the total burst duration $T_{\max}$ is taken as an input parameter. That constraint is represented by determining the upper bounds having control upon $T_{\max}$. These upper bounds can be selected such as $T_r \times (\boldsymbol{GR} + p)$, where $\boldsymbol{GR}$ is a vector consisted of marks (positive real numbers) of length N. We select $\boldsymbol{GR}$ by choosing a GR. p is a positive user-predefined integer satisfying $p \ge N$, and $T_r \ge 2t_b$. For instance, the main rational for choosing the GR in calculating $\boldsymbol{ub}$ of (20) is due to assuming that

$$\left[\boldsymbol{a}_{\mathrm{adaptive}}(\boldsymbol{x})\right] = \underset{\boldsymbol{a}}{\operatorname{argmin}} \left\{ \sum_{n=0}^{N-1} |a_n|^4 |x_n|^2 \int_{-t_b}^{+t_b} \int_{-1/t_b}^{1/t_b} \left| \chi_{s_n s_n} \right|^2 \mathrm{d}\tau \mathrm{d}f_{\mathrm{d}} \right\}. \tag{19}$$

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Table 1 Results of adapting the WVTSC burst

N	$\boldsymbol{a}_{\text{adaptive}}$
4	[0.4419 0.5433 0.4299 0.5699][a]
4	[0.3162 0.5487 0.3162 0.7064][b]
5	[0.3518 0.5264 0.4618 0.4832 0.3904][a]
5	[0.3162 0.6645 0.3759 0.4660 0.3162][b]
6	[0.3377 0.4876 0.4234 0.3335 0.3356 0.4809]
7	[0.3671 0.2962 0.4735 0.3296 0.4638 0.3638 0.3117]
8	[0.3111 0.4042 0.2737 0.3778 0.2641 0.4194 0.2885 0.4397]
9	[0.2498 0.4146 0.2503 0.3274 0.2146 0.3725 0.3594 0.3804 0.3704][a]
9	[0.1371 0.5613 0.1399 0.2568 0.1012 0.3977 0.3561 0.3947 0.3601][b]
10	[0.2704 0.3899 0.2340 0.3256 0.4006 0.2080 0.3944 0.3279 0.2401 0.2993]

[a]The remaining data are resulting from considering all constraints.

[b]Withdrawing the constraint $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$.

the optimal solution might exist in the vicinity of the values of Golmob marks. However, we do not restrict the optimisation model to have GR-based upper bounds. The designer is free to use other criteria for that. However, one can notice that both p and T_r really control $T_{\max}$. On the other hand, the lower bounds $\boldsymbol{lb}$ of (20), as another input to the GA, can be selected to be $[0, T_r, 2T_r, 3T_r \ \ldots, (N-1)T_r]$. However, assuming the initial distribution of time marks based on the GR might help, but it technically implies a difficulty to assume reasonable $\boldsymbol{ub}$ of (20). This assumption requires a broader analytical study, which is outside the scope of this paper.

The weighting coefficients $\{\omega_s\}_{0 \leq s \leq N-1}$ generally quantify how certain the solution of the GA is really about the fact. It is worth noting that the units of the fitness functions $\{F_s\}_{0 \leq s \leq N-1}$ are quite similar. Each one of them represents volume under the ambiguity surface. Hence, we can follow the traditional weighting coefficient settings following [27]. This setting results in moderate weight between each fitness function F_s. Thus, we assume that each ω_s is positive and satisfied to be from 0 to 1. Actually, (20) solution does not change if all weights $\{\omega_s\}_{0 \leq s \leq N-1}$ are multiplied by a constant. Hence, it is the usual practice to choose weights such that $\sum_{s=0}^{N-1} \omega_s = 1$. Equation (20) solution is Pareto-optimal since $\{\omega_s\}_{0 \leq s \leq N-1}$ are positive for each objective F_s [27]. Mathematically speaking, one may find numerous interesting theorems clarifying the relationship between the optimal solution of an optimisation problem to true Pareto-optimal solutions in classical texts, e.g. in [28]. The Pareto-optimal set for the N-objectives problem (20) is non-convex. We have N objectives F_s, and we will follow the vector evaluated approach (VEA). The VEA is used as a weight-based approach for MOO. Comparing with another approach, e.g. the sharing function approach, VEA is better in the sense of computing complexity since we do not require pair-wise distance metric [27]. Moreover, keeping any additional variable for the weight vectors is not needed. To implement (20), we independently apply GA to each subpopulation. Hence, an adequate subpopulation size is required for finding the optimal solution corresponding to each ω_s [27, Sect 5.6.2].

After all, we propose here a sequential optimisation (first weights, then time marks), while the joint optimisation problem is not considered. The result of such probable joint optimisation might be another optimal solution. At first glance, the joint (dual) optimisation process, over the near and far regions of the AF, seems to be very difficult. As a potential way to deal with this issue, we suggest to add optimisation objective representing the region Ω_τ enclosed by the all side lobes of the complete ACF $\chi_{\text{MF}}(\tau, 0)$. This means to minimise a quantity $F_1 = \int_{\Omega_\tau} |\chi_{\text{MF}}(\tau, 0)| \text{d}\tau$ over a set $\boldsymbol{C}$ that represents WVTSC parameters to be optimised ($\boldsymbol{a}$ and $\boldsymbol{T}$), and corresponds to the sub-regions of interest. Then, by implementing just GA to solve the dual case, we aggregate the objective functions F_1, $f(\boldsymbol{a})$, and $F(\boldsymbol{T})$ in one objective optimisation, i.e. $F_{\text{dual}} = \mu_1 \text{F}_1(\boldsymbol{a}, \boldsymbol{T}) + \mu_2 f(\boldsymbol{a}), \ + \mu_2 F(\boldsymbol{T})$, where μ_1, μ_2, and μ_3 are the GA weight coefficients. It is obvious that the unit of both F and f is volume, whereas the F_1 unity is area. As a consequence, the traditional weighting coefficient method results in immoderate weight between each sub-objective function of F_{dual}. Thus, by following [29], one can implement an average adaptive weight approach to achieve global GA search and get better the population diversity of the MOO of F_{dual}. However, the joint (dual) optimisation issue is left for readers to be further researched.

5 Results and discussion

5.1 Basic considerations

To check up our approach, we performed various simulations on a PC with a 2.40 GHz i7-5500U CPU and 8.00 GB installed random-access memory, and using MATLAB© R2014a platform. We will show the simulation results in this section. These results represent the performance of the developed adaptation and optimisation models of the WVTSC waveform design. We will discuss those results successively. We will consider a normal Costas waveform with pulse width of $T = 7\,\mu$s, and the lengths of hopping sequence of Costas code such that $4 \leq N \leq 10$, where $N \leq 3$ is taken as a trivial case. Then, we construct the WVTSC burst model and then simulate its NAF. Rather than writing down the exact code of both the Costas hopping sequence and the GR for each N, we select those codes using a MATLAB script during the simulation.

5.2 Results of the adaptation model

The purpose of the adaptation model is to reduce the SLL in the main lobe area of the NAF at both zero Doppler cut and zero delay cuts. We initialised the WVTSC waveform parameters corresponding to (4) such that $T_r = 3t_b$, $T_n = nT_r$, $a_{\text{initial}}(n) = 1/\sqrt{N}$, and $n = 0, 1, 2, \ldots, N-1$. N weights were adapted to N components of the vector $\boldsymbol{x}$ obtained from Rayleigh distribution. We used the MATLAB functions `makedist` to create Rayleigh probability distribution object, and the function random to initialise the minimisation algorithm with vector $\boldsymbol{x}$. We also used the constrained non-linear problem solver of MATLAB platform fmincon. This solver is usually used to solve any minimisation problem similar to (22). It also implements the interior-point algorithm we considered.

Table 1 show the resulted adaptive weights. Each $\boldsymbol{a}_{\text{adaptive}}$ vector corresponds to $\boldsymbol{x}$ vector realised for each value of N. We considered $\varepsilon_1 = 0.01$ and $\varepsilon_2 = 0.005$. Regarding the resultant ratio quantity RQ for the problem (15), it was much closer to zero, as expected. For example, $RQ = 7 \times 10^{-6} \ll 1$ for $N = 9$, which means that the adaptation model has a strong ability to redistribute the signal energy. Hence, the amplitude of the received signal is almost constant over the time duration of reception. Moreover, we notice from Table 1 that each $\boldsymbol{a}_{\text{adaptive}}$ satisfies all the constraints of (15) and the corresponding upper and lower bounds.

In terms of the processing time, the elapsed times to get each $\boldsymbol{a}_{\text{adaptive}}$ for a single WVTSC withdrawing $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$ when $N =$ 4, 5, 6, 7, 8, 9, and 10, were 0.021, 1.098, 1.440, 3.560, 5.501, 6.925, and 8.270 s, respectively. When $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$ was included, the elapsed times were 1.021, 3.825, 5.032, 7.125, 8.985, 10.562, and 12.085 s, for the same values of N, respectively. Comparing to a full train of L adaptive pulses, the issue of adaptation run-time becomes linked to other practical considerations. For example, whether a single PRF is considered or multi-PRF, and what the value of PRF is used (low, medium, or high), and if the TIR estimation (i.e. $\boldsymbol{x}$) changes between the pulses in one train or not. In addition, those processing times depend on the solver used and how it implements the interior-point algorithm. Our MATLAB code could be optimised to adapting WVTSC bursts dynamically in real time. The interior-point algorithm could be also developed to run much quicker. Whether the search for adaptive weights can be implemented on the fly or not would depend on the infrastructure available in terms of the computation power.

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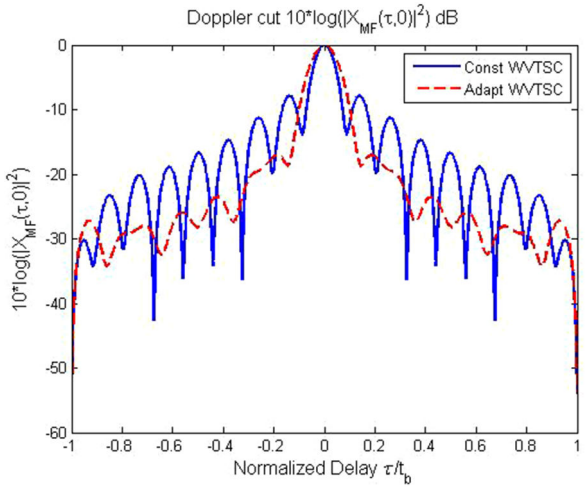


Fig. 2 *ACF in the main lobe area, both constant and adaptive weights are applied, assuming constant time spacing, for* $N = 9$*, withdrawing* $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$

Table 2 SLL characteristics in the main lobe area of the ACF of the WVTSC burst

N	Average sidelobe, dB		Peak sidelobe (PSL), dB	
	Initial	Adapted	Initial	Adapted
4	−9.44	−13.21[a], − 14.19[b]	−8.18	−11.75[a], − 21.92[b]
5	−10.28	−15.27[a], − 15.45[b]	−8.70	−12.54[a], − 21.17[b]
6	−10.96	−16.38	−9.28	−13.00
7	−11.62	−17.74	−8.93	−13.42
8	−12.26	−18.79	−10.53	−16.66
9	−12.06	−19.00[a], − 18.00[b]	−7.89	−13.42[a], − 17.08[b]
10	−12.70	−19.87	−9.92	−16.18

[a]The remaining data are resulting from considering all constraints.

[b]Withdrawing the constraint $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$.

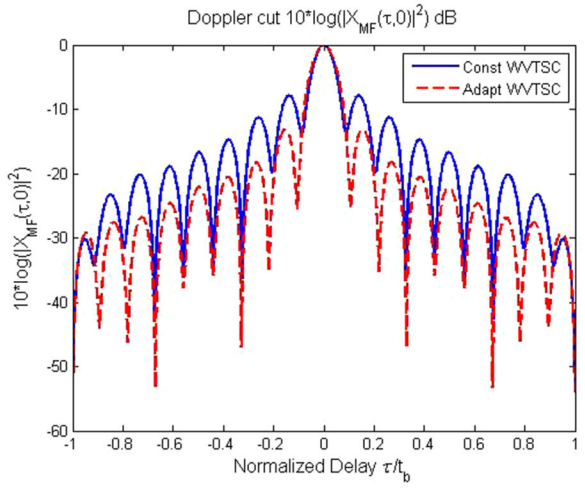


Fig. 3 *ACF in the main lobe area, both constant and adaptive weights are applied, assuming constant time spacing, for* $N = 9$*, including* $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$

Otherwise, it may be done off line, e.g. using real returns data of commercial target models recorded. In any case, the results of the adaptation strategy seem to be technically reasonable and promising, as we will show next.

Fig. 2 shows the NAF at the zero-Doppler cut (i.e. ACF characteristic) of WVTSC burst before and after adaptation, for N=9 withdrawing the constraint $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$. In comparison between the initial (constant) and adaptive weights of WVTSC, the SLLs in the main lobe area of the ACF showed considerable improvement with adaptive weights. The resulted SLLs are too low to mask the main lobe of other possible targets. Hence, the much better ability of the adaptive WVTSC was obtained to detect weaker targets whose main lobe is close to that one of the targets under study. Moreover, Fig. 2 indicates that the delay resolution has changed a little bit. Actually, this change is not important in the sense of the nominal delay resolution defined as the width of the main lobe at −3 dB of the ACF [5].

Table 2 represents the SLL of ACF characteristic of the WVTSC waveform for $4 \leq N \leq 10$ before and after adaptation procedures. The column 1 of Table 2 represents the average ACF SLLs of adaptive WVTSC waveform. The 'average sidelobe' metric was recently used in [14]. The column 2 shows the resultant PSL of the waveform ACF. Both columns confirm the validity of the adaptation model on designing WVTSC signals. For example, we achieved over 4–9 dB in reducing the PSL of ACF and about 4–7 dB in average ACF sidelobe reduction for $N = 9$. In this study, we do not generally have interest in the integrated side-lobe ratio (ISLR) metric, because we do not focus on applying the WVTSC model in synthetic-aperture radar (SAR) imaging. However, let us consider that the ISLR is the ratio of the total energy contained in side lobes to the energy contained in just the main lobe (peak response) of the complete ACF. Then, let us apply the same setting for $N = 9$ along with all the constraints suggested in both (15) and (20). As a result, the ISLR of the initial WVTSC would be −9.9844 dB. The ISLR of the optimised WVTSC would be −19.9694 dB. On the other hand, if one withdraws $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$ from (15), the ISLR of the optimised WVTSC would be −20.4809 dB. The results of the remaining N cases generally imply an average ISLR gain of 11 dB. Therefore, there is also a considerable improvement in the ISLR point of view. Nevertheless, we cannot claim the usefulness of the proposed WVTSC in SAR imaging until further researches should be done according to the application.

Fig. 3 shows the ACF resulted from solving (15) with the same setting of Fig. 2, including the constraint $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$. In general, considering $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$ is up to the designer and depends on the transmitting power operation requirements. However, considering that non-linear constraint or does not really influence the main purposes of our approach. We insured that by performing several simulations. For example, withdrawing this constraint results in much more reduction in PSL. This is obvious by comparing Figs. 2 and 3, and using Table 2 data. For instance, when $N = 5$, if we withdraw that constrain, the PSL gain is about 12.5 dB. If we include it, the PSL gain is just about 4 dB. However, we found that withdrawing $\text{var}(\boldsymbol{a}) \leq \varepsilon_2$ is the reason behind that little bit change in the delay resolution mentioned above. From Fig. 3, it is clear that there is no change in the delay resolution. Therefore, if we are interested to have no change in the target range resolution, we need to consider all constraints of (15). Moreover, the reductions in average of ACF SLLs in the main lobe area are quasi-similar. This is clear by comparing the result of Fig. 2 settings with that of Fig. 3, and using Table 2 data (e.g. $N = 4, 5, 9$).

Fig. 4 shows the normalised NAF at zero delay cut resulting from solving (15) with the same setting of Fig. 3, including all the constraints. As is expected, there is no change in the nominal Doppler resolution. In addition, it is obvious that the zero delay cut of adaptive WVTSC shows a considerable improvement in the SLL surrounding the main lobe area compared with the initial WVTSC. Numerically, for $N = 9$, the PSL gain was about 8.17 dB. We also achieved about 9.82 dB in average sidelobe reduction gain compared before and after adaptation. Table 3 presents the SLL of the delay cut characteristic of the WVTSC waveform, for $4 \leq N \leq 10$ before and after adaptation. The column 1 presents the average side lobes resulted of adaptive and optimised WVTSC, while the column 2 shows the resultant PSL. In the simulation, we ran just the algorithm of (15). However, if (16) or (19) is used instead of (15), we found that there was no important change in the main results in Figs. 2 and 3. Generally, (16) differs from (15) in considering a desired AF or not. The complexity of (15) depends on the shape of the desired AF and the corresponded Ω. If that shape is more complex, (16) has less complexity, for the same Ω. Equation (19) has less complexity than both (15) and (16). If we use (19), just the main lobe part of the AF is considered. Thus, (19) can be chosen if the power of computing is low.

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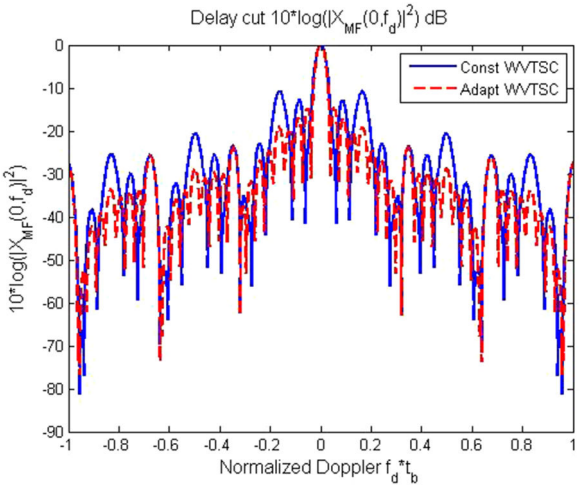


Fig. 4 *Normalised NAF at zero delay cut in main lobe area, both constant and adaptive weights are applied, and constant time spacing is assumed, for $N = 9$, considering all the constraints of (15)*

Table 3 Delay cut SLLs in the main lobe area of the WVTSC burst considering all constraints

N	Average delay cut sidelobe, dB		Delay cut peak sidelobe (PSL), dB	
	Initial	Adapted	Initial	Adapted
4	−13.27	−20.43	−11.20	−15.30
5	−14.27	−21.84	−8.35	−12.14
6	−15.07	−23.52	−11.23	−14.05
7	−15.81	−24.55	−9.95	−12.64
8	−16.53	−25.23	−15.67	−21.47
9	−16.14	−25.98	−10.8	−18.97
10	−16.85	−26.54	−11.84	−20.04

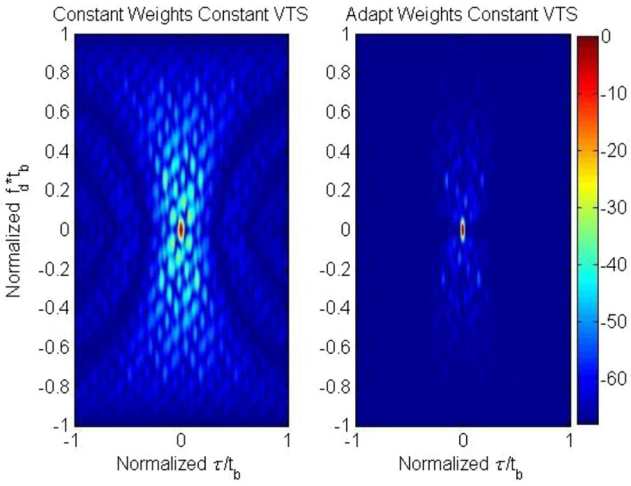


Fig. 5 *Full 2D map (both delay and Doppler dimensions) of the normalised NAF in main lobe area, constant and adaptive weights are applied, where constant time spacing is applied, for $N = 9$, considering all the constraints of (15), and a log colour bar scale is used for the two parts to better represent the SLL*

To more clarify the SLL improvement, Fig. 5 presents that improvement by a full 2D map (both delay and Doppler dimensions) of the normalised NAF, considering adaptive and initial WVTSC. It is important to state here that choosing the Ω actually controls the computing complexity as the minimisation process progresses. The more the Ω is scanning the entire delay-Doppler plane, the more accurate the adaptation process and much better improvement in the NAF. That requires high powerful computational resources, but the results will not change technically. Moreover, when the statistics of $\boldsymbol{x}$ is known, the adaptation procedure can be extended, but the maximum likelihood estimation of $\boldsymbol{x}$ should be done. However, the designer must carefully choose the value of ε_2 corresponding to the transmitting power specifications, because $\boldsymbol{a}_{\text{adaptive}}$ depends on it. On the other hand, if a desired AF is assumed with a thumbtack profile with a finite main-lobe width, the main results of this work, do not considerably change, except that the run time might be larger. The same is for another shape of the desired AF.

Table 4 Optimised VTS marks of the WVTSC burst

N	$\boldsymbol{T}_{\text{Optimal}}$, μs
4	[0 11.95 22.94 52.44]
5	[0 14.98 20.87 39.67 59.22]
6	[0 4.77 8.27 13.35 32.55 77.51]
7	[0 4.46 7.31 11.35 23.17 42.14 80.75]
8	[0 4.07 7.03 9.64 12.10 28.37 77.78 99.36]
9	[0 2.83 6.26 9.51 16.31 22.39 54.30 58.90 103.79]
10	[0 2.78 6.50 9.50 17.95 25.53 29.09 44.56 69.87 98.94]

5.3 Results of the optimisation model

The purpose of the optimisation model is to reduce the SLL in the recurrent sidelobe area of the NAF at zero Doppler cut using optimal VTS. We initialised the WVTSC VTS parameters such that $T_n = nT_r$. We also initialised the WVTSC with $\boldsymbol{a}_{\text{adaptive}}$. N parameters (i.e. $\{T_n\}_{0 \le n \le N-1}$) were optimised using (20). We used the constrained linear problem solver of MATLAB platform, gamultiobj. This solver is used to get the solution of any MOO problem. It actually implements the *multi-objects GA* based on evolutionary algorithms. Table 4 illustrates the resultant optimised VTS marks of the WVTSC waveform for $4 \le N \le 10$ considering all the constraints of (15). We notice from Table 4 that each vector $\boldsymbol{T}_{\text{Optimal}}$ found satisfies the linear constraints of (20) and the corresponding upper and lower bounds.

In terms of the processing time, we found that the elapsed times to get each $\boldsymbol{T}_{\text{Optimal}}$ for a single WVTSC, including just three recurrent sidelobe functions, were 7.814, 8.905 , 9.127, 10.913, 11.632, 12.097, and 13.278 s for $N = 4$, 5, 6, 7, 8, 9, and 10, respectively. When all $N-1$ recurrent sidelobe functions are included in (20), it took 7.546, 13.957, 15.964, 18.157, 19.970, 22.737 , and 25.485 s. In comparison with a full train of L optimised WVTSCs, the run-time issue of the VTS optimisation becomes linked to the same considerations such as those regarding to the adaptation model. In addition, that issue becomes linked to the constraints on $T_{\max}$ considered. We cannot tend to consider the practical implementation of the described optimisation feasible as easy. However, we notice the usefulness of reducing the number of objective functions F_s for (20). The run-time of three objective functions is obviously shorter. That is up to the designer and depends on the infrastructure available and requirements. The results were obtained considering that the maximum number of generations in GA algorithm was about 1000. The search time depends on the MATLAB solver used and how it implements the GA. Thus, our code needs to be optimised to run in real-time, considering it is still an un-modified engineering application for now. Otherwise, the search could be done offline. Anyway, developing an optimised code with less generations and using both parallelisation and vectorisation programming schemes, the GA evolutionary approach can identify optimal solutions very quickly indeed. The parallelisation scheme can be implemented on a single multi-core machine, such as the computer used. However, the proposed optimisation strategy for the WVTSC seems to be also reasonable and technically sound, as we will show next.

We notice from Table 4 that, even if $T_{\max} \gg T$ for all N (e.g. for $N = 9$, $T_{\max} = T_{N-1} + t_b = T_8 + t_b = 110.79\,\mu\text{s} \simeq 15T$) the pulse compression gain results in as same as that of the original Costas waveform. This result is natural since the effective transmission duration of the proposed WVTSC burst is still equal to $T = Nt_b$, but not $T_{\max}$ and the BW of the WVTSC is unchanged. Despite

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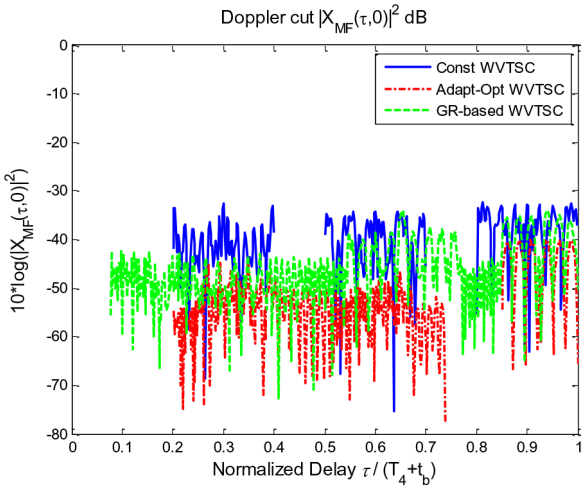


Fig. 6 *First three recurrent sidelobes of the normalised NAF at zero Doppler cut in the sidelobe area ($T_4^{\text{optimal}} - t_b \le \tau \le T_4^{\text{optimal}} + t_b$, but just positive values are illustrated), where constant, optimised, and Golomb-ruler-based time spacing marks are applied, and adaptive weights are assumed, for N=9*

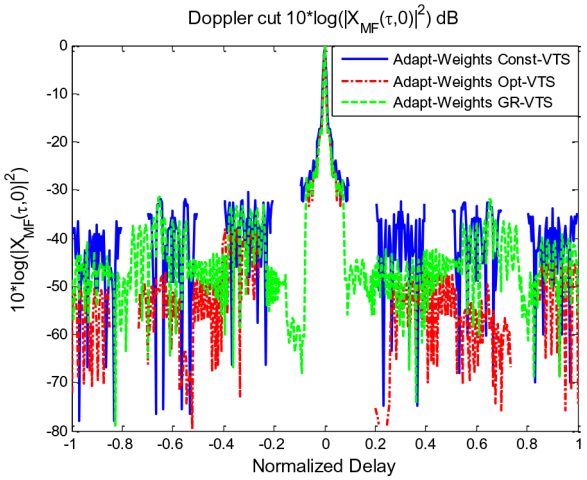


Fig. 7 *Quasi-complete ACF (considering both the main lobe part and the sidelobe part) of: 'const VTS', ' Optimal VTS', 'Golomb ruler-based VTS', adaptive weights are assumed, for $N = 9$, and the delay is normalised by $T_4^{\text{optimal}} + t_b$*

Table 5 Average of $N - 1$ recurrent SLLs of NAF at zero Doppler cut for initial and optimised WVTSC

N	Average recurrent sidelobe, dB	
	Initial	Optimised
4	−33.54	−45.21
5	−35.57	−46.15
6	−36.29	−44.40
7	−38.63	−47.82
8	−39.48	−47.45
9	−39.62	−48.33
10	−40.44	−52.82

there is no improvement in the pulse compression gain, for small N, the WVTSC model yields a considerable improvement in SLL without needing to assume a higher value of N. However, if a larger pulse compression gain is required, a higher N can be used, knowing that the WVTSC optimisation can be extended for any value of N. Moreover, using another $\{f_n\}$ distribution rather than the Costas method so that BW can be larger supposing δf could optimally change rather than being fixed, then larger pulse compression gain can be achieved without increasing N. This might refer to an application that requires wider BW. This will correspond to a variable frequency spacing similar to the proposed VTS. This solution, like our model, involves a new way of thinking in dealing with the frequency and time domains of the waveform. Certain methods of distributing frequencies and the transmission moments of the burst pulses could be considered. Common frequency and phase codes can be considered as distribution, but no one has thought of what the distribution method it should be better defined in the sense of the distribution of τ and f_{d} in a prior interval of interest. This work appears to be the first step with regard to its new methodology of radar waveform design thinking.

Fig. 6 shows the first three recurrent side lobes of the NAF at zero Doppler cut considering both constant (initial) and optimised VTS. Fig. 6 also shows the result WVTSC based on the GR method. Fig. 6 confirms the validity of the optimised WVTSC. It is so obvious that recurrent SLLs for $T_s^{\text{optimal}} - t_b \le \tau \le T_s^{\text{optimal}} + t_b$, are too low than those for $sT_r - t_b \le \tau \le sT_r + t_b$, as expected. Fig. 6 also shows the ACF performance of WVTSC based on the GR distribution as an intermediate performance case, compared with constant and optimised VTS. Therefore, we can say that GR-based VTS is not an optimal solution by means of decreasing the SLL. It instead just gives good performance. Moreover, it is clear from Fig. 6 that zeros occurring between recurrent side lobes are reduced or removed as expected. Hence, the recurrent SLLs are improved to be too low to mask the main lobe.

Fig. 7 shows a quasi-complete ACF (three recurrent sidelobe functions). We notice from Fig. 7, as expected, that there is no important impact of adaptive weights on recurrent SLLs for both constant VTS and GR-based VTS.

Table 5 presents average $N - 1$ recurrent SLL of zero Doppler cut for $4 \le N \le 10$ before and after optimisation procedure. It is obvious that $\boldsymbol{T}_{\text{Optimal}}$ yields much better SLL. For example, about 9 dB was a considerable gain in average SLL for $N = 9$. We also found, for any N, that using $\boldsymbol{T}_{\text{Optimal}}$ as an input to $\chi_{\text{MF}}^{(\text{ml})}(\,.\,)$ will not

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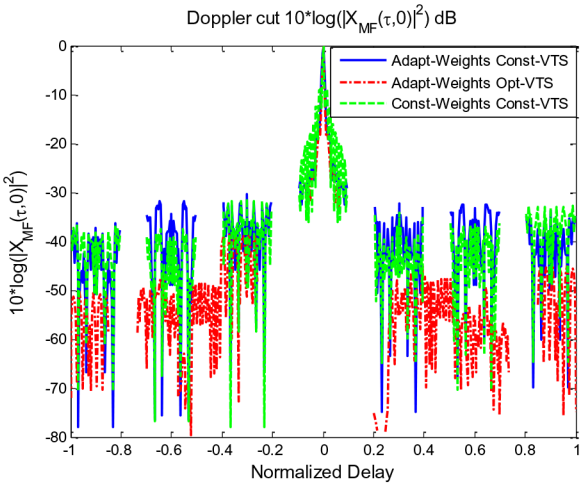


Fig. 8 *Comparing the three cases of the quasi-complete ACFs (considering both the main lobe part and three sidelobe parts) of 'const WVTSC', 'optimised weights + constant time marks', 'optimised weights and optimised time marks', for $N = 9$, and the delay is normalised by $T_4^{\text{optimal}} + t_b$*

change the SLL in the main lobe area, comparing with initial VTS. This result is natural and expected. It gives an added value to our design and shows the powerful of separating the mathematical formula of NAF into two parts, i.e. (see (12)).

Fig. 8 shows a comparison between three cases of ACF (effects of the different optimisation steps). The first uses constant weights and constant VTS marks. The second uses adaptive weights and constant VTS. The third refers to sending adaptive weights with optimal VTS. It is obvious, as expected, that the adaptive weights do not influence the recurrent SLLs of ACF. From Fig. 8, we notice that those levels are approximately the same. Hence, our aim of achieving a considerable reduction in recurrent SLLs was obtained by using optimal VTS.

Finally, we present the results of the Doppler intolerance issue with regard to the WVTSC model. Firstly, Let us define the degree of Doppler intolerance of the optimised WVTSC $B_{\text{d}}^{\text{eff}}$ as an effective Doppler BW taken at -3 dB of the NAF. We found that $B_{\text{d}}^{\text{eff}}$ was $0.036/t_b$, $0.030/t_b$, $0.025/t_b$, $0.020/t_b$, $0.018/t_b$, $0.016/t_b$, $0.014/t_b$, for $N = 4, 5, 6, 7, 8, 9$, and 10, respectively. For instance, we notice that, for $N = 9$, the optimised WVTSC yields an effective Doppler BW of about 45 kHz. In the X-band, this value is hardly sufficient to detect very fast-moving targets, e.g. fighter planes. Thus, Doppler intolerance mitigation may be required for the WVTSC model. In Section 3.2, we have suggested three solutions to the Doppler intolerance problem. One of them is to consider $\chi_{\text{Opt}}(.)$ shape as a constant amplitude ridge along the zero delay in a desired Doppler BW. This requires added constraints yielding high-computational cost added at the transmitter side. For us, Doppler compensation technique proposed in [24] is more attractive since there are Doppler phase shifts introduced by the two functions $\kappa_n(\tau, f_{\text{d}})$ and $\zeta_{n_1 n_2}(\tau, f_{\text{d}})$ in our model. In addition, it could be processed at the receiver side.

6 Conclusion

In this article, we proposed a new burst-like waveform based on Costas code such that it had different weights corresponding to the frequency subcarriers used. We incorporated target scattering coefficients into the received replica of the signal. The distribution of the proposed waveform pulses has been also modified by adding VTS between the pulses. We named the resultant waveform as a WVTSC burst that is no longer called a Costas waveform. We computed the analytic expressions of the AF of the WVTSC waveform. We developed an adaptive and optimal WVTSC design. The weights of the proposed burst were adapted to target scattering coefficients. The VTS between its pulses was optimised. Both interior-point algorithm and GA were applied to solve the minimisation problems. The simulation results showed a considerable improvement in the SLL of the AF in both the main lobe area and the recurrent sidelobe areas. This improvement was verified by comparing the SLL of the optimised WVTSC with that of a Costas waveform having constant weighted and constant time spacing, and with that of GR-based VTS burst. This SLL reduction in the developed waveform was achieved without the need to increase the Costas code length. The new design waveform shown has no important impact on the original delay-Doppler resolution. Hence, we achieved much better characteristics of the AF. The WVTSC could be useful in the staggered PRF radar. The WVTSC could be generalised by using other methods of distributing both frequencies and the transmission moments of a burst-like waveform. The proposed adaptation and optimisation models can be also extended for those generalised burst-like waveforms. We discussed the Doppler intolerance issue of the WVTSC. We suggested useful solutions to mitigate the intolerance effect. One of them is to use the compensation technique at the receiver side. This solution was recently developed and may be the direction to future work. We may also apply variable frequency spacing between the WVTSC subcarriers such that the resultant frequency separation optimally changes rather than being fixed. We would then verify the performance of the optimisation model through simulation or field experiments.

7 Acknowledgments

We cannot express enough thanks to anonymous reviewers in the IET-RSN for their valuable comments on this article. Without their comments and suggestions, the article writing would not have been revised for the better. We also want to thank Professor Nidal Chamoun for his valuable advice in developing the article language.

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9. Appendix

We can write (9) such that

$$\begin{aligned}\chi_{\mathrm{MF}}(\tau, f_{\mathrm{d}}) &= (t_b/T_{\max})\\ &\left\{\frac{1}{t_b}\sum_{n=0}^{N-1} x_n^*|a_n|^2\int_{-\infty}^{+\infty} s_n(t-T_n)s_n^*(t-\tau-T_n)\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t\right.\\ &+\frac{1}{t_b}\sum_{n_1=0}^{N-1}\sum_{\substack{n_2=0\\ n_2\neq n_1}}^{N-1} x_{n_2}^* a_{n_2}^* a_{n_1}\int_{-\infty}^{+\infty} s_{n_1}(t-T_{n_1})\\ &\left.s_{n_2}^*(t-\tau-T_{n_2})\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t\right\}\end{aligned} \tag{23}$$

Now, defining for $n_1 = n_2 = n$, the function

$$\Phi_{nn}(\tau, f_{\mathrm{d}}) = (1/t_b)\times\int_{-\infty}^{+\infty} s_n^*(t-T_n)s_n(t-\tau-T_n)\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t, \tag{24}$$

and for $n_1 \neq n_2$ the function

$$\Phi_{n_1n_2}(\tau, f_{\mathrm{d}}) = (1/t_b)\times\int_{-\infty}^{+\infty} s_{n_1}^*(t-T_{n_1})s_{n_2}(t-\tau-T_{n_2})\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t, \tag{25}$$

We notice that

$$\Phi_{nn}^*(\tau, f_{\mathrm{d}}) = (1/t_b)\times\int_{-\infty}^{+\infty} s_n(t-T_n)s_n^*(t-\tau-T_n)\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t, \tag{26}$$

$$\Phi_{n_1n_2}^*(\tau, f_{\mathrm{d}}) = (1/t_b)\times\int_{-\infty}^{+\infty} s_{n_1}(t-T_{n_1})s_{n_2}^*(t-\tau-T_{n_2})\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t\,. \tag{27}$$

Changing the variable in (26) such that $\sigma = t - T_n$, we get

$$\begin{aligned}\Phi_{nn}(\tau, f_{\mathrm{d}}) &= \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_n}\frac{1}{t_b}\int_{-\infty}^{+\infty} s_n^*(\sigma)s_n(\sigma-\tau)\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}\sigma}d\sigma\\ &= \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_n}\phi_{nn}(\tau, f_{\mathrm{d}})\,,\end{aligned} \tag{28}$$

where we put

$$\phi_{nn}(\tau, f_{\mathrm{d}}) = \frac{1}{t_b}\int_{-\infty}^{+\infty} s_n^*(\sigma)s_n(\sigma-\tau)\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}\sigma}\mathrm{d}\sigma\,. \tag{29}$$

In fact, (29) represents the autocorrelation function of the sub-pulse $s_n(\sigma)s_l(t)$, and can be simplified directly by using the Costas calculation corresponding to (21) of his original paper [10], so we get for $|\tau| \leq t_b$

$$\phi_{nn}(\tau, f_{\mathrm{d}}) = \frac{(t_b-|\tau|)}{t_b}\mathrm{sinc}(\pi f_{\mathrm{d}}(t_b-|\tau|))\mathrm{e}^{[\mathrm{j}\pi f_{\mathrm{d}}(t_b+\tau)-\mathrm{j}2\pi f_n\tau]}, \tag{30}$$

and zero elsewhere. Substituting (30) into (28), we get

$$\Phi_{nn}(\tau, f_{\mathrm{d}}) = \frac{(t_b-|\tau|)}{t_b}\mathrm{sinc}(\pi f_{\mathrm{d}}(t_b-|\tau|))\mathrm{e}^{[\mathrm{j}\pi f_{\mathrm{d}}(2T_n+t_b+\tau)-\mathrm{j}2\pi f_n\tau]}, \tag{31}$$

for $|\tau| \leq t_b$, and zero elsewhere. If we put $\kappa_n(\tau, f_{\mathrm{d}}) = -f_{\mathrm{d}}(2T_n + t_b + \tau) + 2f_n\tau$, and then substituting (31) into (26) we get for $|\tau| \leq t_b$

$$\begin{aligned}&\frac{1}{t_b}\int_{-\infty}^{+\infty} s_n(t-T_n)s_n^*(t-\tau-T_n)\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t = ((t_b-|\tau|)/t_b)\mathrm{sinc}\\ &(\pi f_{\mathrm{d}}(t_b-|\tau|))\mathrm{e}^{\mathrm{j}\pi\kappa_n(\tau, f_{\mathrm{d}})},\end{aligned} \tag{32}$$

and zero elsewhere. Then, substituting (32) into the first term of (23) and for $|\tau| \leq t_b$ we get the main lobe part of AF

(see (33))

and zero elsewhere. Hence, (33) corresponds to the required (13).

On the other hand, by changing the variable in (25) such that $\sigma = t - T_{n_1}$ remembering in this case that $n_1 \neq n_2$, we get

(see (34))

Again, changing the variable in (34) such that $\tau' = \tau - (T_{n_1} - T_{n_2})$, we get

$$\begin{aligned}\Phi_{n_1n_2}(\tau' + (T_{n_1}-T_{n_2}), f_{\mathrm{d}}) &= \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_{n_1}}\\ &\times\left(\frac{1}{t_b}\int_{-\infty}^{+\infty} s_{n_1}^*(\sigma)s_{n_2}(\sigma-\tau')\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}\sigma}\mathrm{d}\sigma\right)\\ &= \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_{n_1}}\phi_{n_1n_2}(\tau', f_{\mathrm{d}})\,,\end{aligned} \tag{35}$$

where we put

$$\phi_{n_1n_2}(\tau', f_d) = \frac{1}{t_b}\int_{-\infty}^{+\infty} s_{n_1}^*(\sigma)s_{n_2}(\sigma-\tau')\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}\sigma}\mathrm{d}\sigma\,. \tag{36}$$

In fact, (36) represents the CCF between two sub-pulses $s_{n_1}(\sigma)$ and $s_{n_2}(\sigma)$ and can be simplified directly by using the Costas calculation corresponding to (18) of his original paper [10], hence we get for $|\tau'| \leq t_b$

$$\chi_{\mathrm{MF}}^{(\mathrm{ml})}(\tau, f_{\mathrm{d}}) = (1/T_{\max})\times\sum_{n=0}^{N-1} x_n^*|a_n|^2(t_b-|\tau|)\mathrm{sinc}(\pi f_{\mathrm{d}}(t_b-|\tau|))\mathrm{e}^{\mathrm{j}\pi\kappa_n(\tau, f_{\mathrm{d}})} \tag{33}$$

$$\Phi_{n_1n_2}(\tau, f_{\mathrm{d}}) = \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_{n_1}}\left(\frac{1}{t_b}\int_{-\infty}^{+\infty} s_{n_1}^*(\sigma)s_{n_2}(\sigma+(T_{n_1}-T_{n_2})-\tau)\mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}\sigma}\mathrm{d}\sigma\right). \tag{34}$$

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$$\begin{aligned}\phi_{n_1n_2}(\tau', f_{\mathrm{d}}) &= \phi_{n_1n_2}(\tau - (T_{n_1} - T_{n_2}), f_{\mathrm{d}}) \\ &= \left(\frac{(t_b - |\tau - (T_{n_1} - T_{n_2})|)}{t_b}\right) \\ &\quad \times \mathrm{sinc}(\pi\alpha(t_b - |\tau - (T_{n_1} - T_{n_2})|)) \\ &\quad \times \mathrm{e}^{[-\mathrm{j}\pi\alpha(\tau + t_b - (T_{n_1} - T_{n_2})) - \mathrm{j}2\pi f_{n_2}(\tau - (T_{n_1} - T_{n_2}))]}\end{aligned} \tag{38}$$

$$\phi_{n_1n_2}(\tau', f_{\mathrm{d}}) = \frac{(t_b - |\tau'|)}{t_b}\mathrm{sinc}(\pi\alpha(t_b - |\tau'|))\mathrm{e}^{[-\mathrm{j}\pi\alpha(t_b + \tau') - \mathrm{j}2\pi f_{n_2}\tau']} \tag{37}$$

,

and zero elsewhere, where $\alpha = f_{n_1} - f_{n_2} - f_{\mathrm{d}}$ is corresponding to (20) in [10]. Substituting (37) into (35) and returning to the variable τ, we get for $|\tau'| \le t_b$

(see (38))

and zero elsewhere. We put $\eta_{n_1n_2}(\tau) = \tau - (T_{n_1} - T_{n_2})$ and $\zeta_{n_1n_2}(\tau, f_{\mathrm{d}}) = \alpha(t_b + \eta_{n_1n_2}(\tau)) + 2f_{n_2}\eta_{n_1n_2}(\tau) - 2f_{\mathrm{d}}T_{n_1}$. Then, substituting (38) into (35), we get for $|\tau'| \le t_b$

$$\begin{aligned}\Phi_{n_1n_2}(\tau', f_{\mathrm{d}}) &= \mathrm{e}^{+\mathrm{j}2\pi f_{\mathrm{d}}T_{n_1}}\phi_{n_1n_2}(\tau', f_{\mathrm{d}}) \\ &= ((t_b - |\eta_{n_1n_2}(\tau)|)/t_b)\mathrm{sinc}(\pi\alpha(t_b - |\eta_{n_1n_2}(\tau)|))\mathrm{e}^{-\mathrm{j}\pi\zeta_{n_1n_2}(\tau}\end{aligned} \tag{39}$$

and zero elsewhere. Substituting (39) into (27), we get

$$\begin{aligned}&\frac{1}{t_b}\int_{-\infty}^{+\infty} s_{n_1}(t - T_{n_1})s^*_{n_2}(t - \tau - T_{n_2})\mathrm{e}^{-\mathrm{j}2\pi f_{\mathrm{d}}t}\mathrm{d}t \\ &= ((t_b - |\eta_{n_1n_2}(\tau)|)/t_b)\mathrm{sinc}(\pi\alpha(t_b - |\eta_{n_1n_2}(\tau)|))\mathrm{e}^{+\mathrm{j}\pi\zeta_{n_1n_2}(\tau, f_d)},\end{aligned} \tag{40}$$

Then, substituting (40) into the second term of (23) and for $|\eta_{n_1n_2}(\tau)| \le t_b$ we get the recurrent sidelobe part of AF as follows:

$$\begin{aligned}\chi^{(\mathrm{sl})}_{\mathrm{MF}}(\tau, f_{\mathrm{d}}) &= (1/T_{\max})\sum_{n_1=0}^{N-1}\sum_{\substack{n_2=0 \\ n_2 \ne n_1}}^{N-1} x^*_{n_2}a^*_{n_2}a_{n_1}(t_b - |\eta_{n_1n_2}(\tau)|) \\ &\quad \times \mathrm{sinc}(\pi\alpha(t_b - |\eta_{n_1n_2}(\tau)|))\mathrm{e}^{\mathrm{j}\pi\zeta_{n_1n_2}(\tau, f_{\mathrm{d}})},\end{aligned} \tag{41}$$

and zero elsewhere. Hence, (41) corresponds to the required (14).

Therefore, we get (12) by substituting (33) and (41) in (23). Finally, Case 1 (i.e. (15)) in Section 4.2 corresponds to (32), while Case 2 (i.e. (16)) corresponds to (40).

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